

Non-Linear Inflation Dynamics In Menu Cost Economies

Andrés Blanco
Atlanta Fed and Emory

Corina Boar
NYU and NBER

Callum Jones
Fed

Virgiliu Midrigan
NYU and NBER

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Overview

- Fraction of price changes *increases rapidly* during inflation surges
 - ⇒ Phillips curves non-linear
- Standard menu cost models predict Phillips curves are linear
 - when consistent with the distribution of micro price changes
 - for moderate inflation rates observed in advanced economies
 - ⇒ Cannot reproduce increase in frequency, so linear Phillips curve
(amplified with strategic complementarities in price setting)

Our Resolution

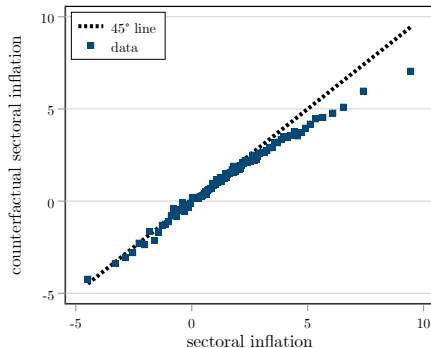
- Extend multi-product menu cost model
 - low elasticity of substitution between products of a firm
 - strategic complementarities at firm, not product, level
- Model implies less within-firm misallocation from price dispersion
 - require smaller menu costs to reproduce distribution of price changes
- Our model predicts non-linear output responses to monetary shocks
 - in contrast to standard models
 - due to strong response in the frequency of adjustment

Motivating Facts

Data

- UK micro-price data underlying construction of CPI
- Decompose $\pi_t(s)$ extensive and intensive margin (Klenow-Kryvtsov, 2005)
- $\pi_t(s) = \Delta_t(s)f_t(s)$
 - $\Delta_t(s)$: average price change conditional on adjustment
 - $f_t(s)$: fraction of price changes
- Isolate role of intensive margin by computing $\pi_t^c(s) = \Delta_t(s)\bar{f}(s)$
 - $\bar{f}(s)$: average frequency in sector s

Evidence From All Sectors



Extensive margin of price adjustment important at high rates of inflation

bread

bread & cereal

Single-Product Model

Model Overview

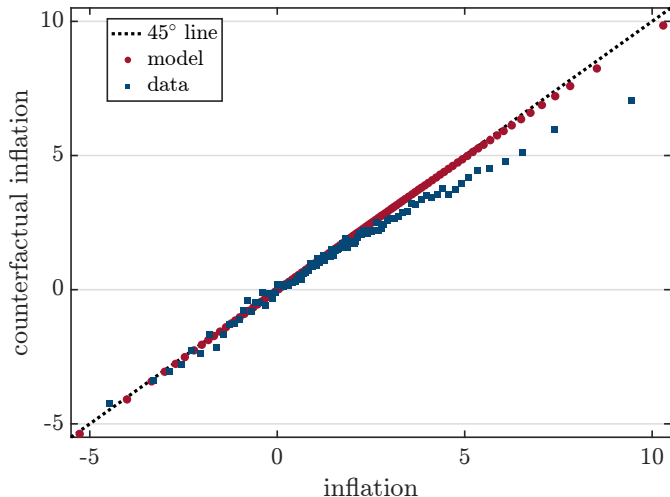
- Continuum of ex-ante identical sectors
- Firms subject to
 - sectoral productivity shocks $e_t(s)$
 - Gaussian idiosyncratic quality shocks $u_t(f, s)$
- Dixit-Stiglitz aggregator for sectoral good output (elasticity σ)
- Produce with decreasing returns (η), labor-only technology
- Random menu costs to match distribution small price changes
- Consumers (β): log-linear preferences + cash in advance constraint
 - nominal wage is equal to money supply
 - firm's problem in a sector independent of aggregate variables

Parameterization

- Assigned
 - period 1 month, discounting ($\beta = 0.99$)
 - firms' demand elasticity ($\sigma = 6$), spawn of control ($\eta = 2/3$)
- Choose menu cost and std. of shocks to match UK data

	Data	Model
std dev. $\pi_t(s)$	0.029	0.029
frequency Δp	0.116	0.116
<i>distribution of Δp</i>		
mean	0.018	0.018
std. dev.	0.188	0.188
kurtosis	3.609	3.609
<i>distribution of Δp</i>		
10 th percentile	0.018	0.021
50 th percentile	0.104	0.118
90 th percentile	0.334	0.315

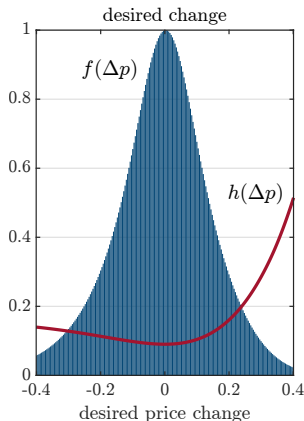
Importance of Extensive Margin



Cannot reproduce extensive margin at high rates of inflation

Frequency of price changes does not move as data

Why is the extensive margin so small?



- Significant menu cost for price adjustment with ...
large probability of zero menu cost to match small price changes

⇒ Small response of frequency/extensive margin

Multi-Product Model

Overview

- Midrigan (11) and Alvarez–Lippi (14) multi-product model
 - firms sell continuum of products
 - product quality shocks $z_{it}(f, s)$, in addition to sector-specific $e_t(f, s)$
 - economies of scope in price adjustment: menu cost $\bar{\xi}$ to change all prices
- Add two ingredients
 - specific factor (e.g. managerial input) mobile across products within firm
 - low elasticity of substitution between products of a given firm

Technology

- Composite good of firm f

$$y_t(f, s) = \left(\int_0^1 \left(\frac{y_{it}(f, s)}{z_{it}(f, s)} \right)^{\frac{\gamma-1}{\gamma}} di \right)^{\frac{\gamma}{\gamma-1}}$$

- Individual varieties produced using labor and specific factor m_{it}

$$y_{it}(f, s) = e_t(s)(f, s) z_{it}(f, s) m_{it}(f, s)^{1-\eta} l_{it}(f, s)^\eta$$

- Specific factor mobile across products, fixed at firm level

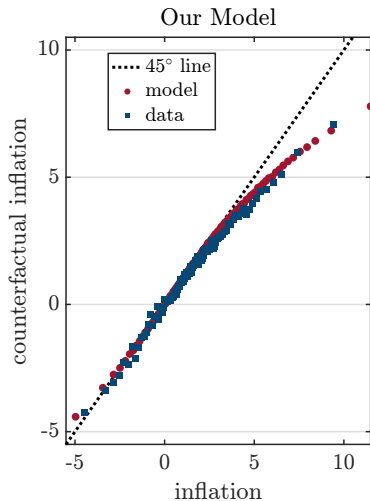
$$\int m_{it}(f, s) di = 1 \quad \text{vs.} \quad m_{it}(f, s) = 1$$

Parameterization

- Assigned: same as earlier
- New parameter: $\gamma = 1$

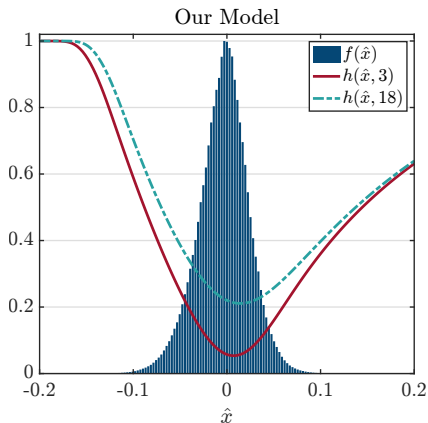
	Data	Our model
std dev. $\pi_t(s)$	0.029	0.029
frequency Δp	0.116	0.116
<i>distribution of Δp</i>		
mean	0.018	0.018
std. dev.	0.188	0.188
kurtosis	3.609	3.873
<i>distribution of Δp</i>		
10 th percentile	0.018	0.020
50 th percentile	0.104	0.115
90 th percentile	0.334	0.311

Importance of Extensive Margin



reproduces the extensive margin at high rates of inflation

Why is the extensive margin so big?



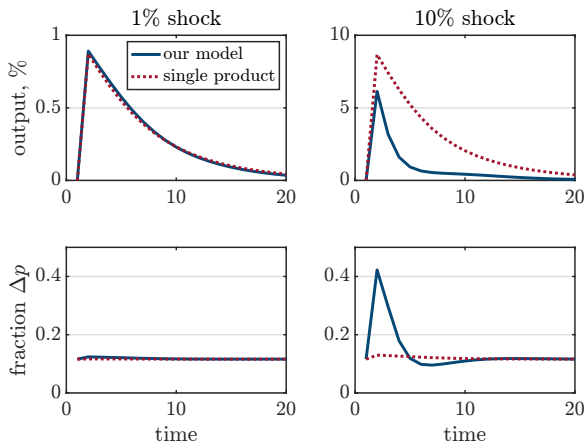
- Small menu cost
 - No probability of zero menu cost
- ⇒ Elasticity of frequency of price change close to data

Real Effects of Monetary Shocks

Impulse Responses to One-Time Shocks

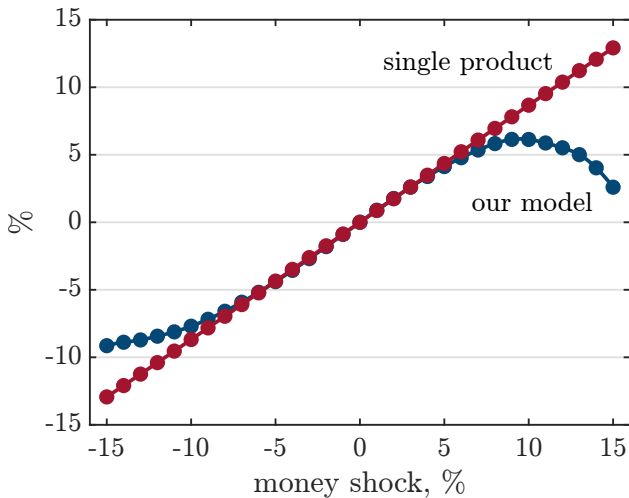
- Compute y_t responses to one-time unanticipated changes in M_t
 - for shocks of different sizes to gauge non-linearity
 - contrast our model to single product
- Since $P_t y_t = M_t$, larger response of y_t due to slower P_t response

Output

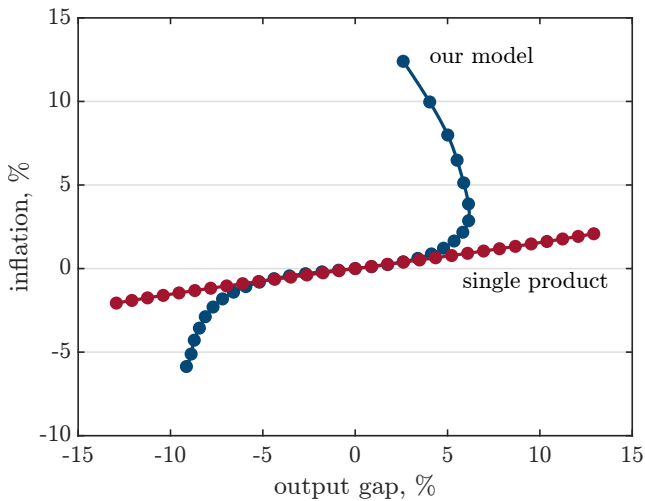


- For small shocks, same IRF
- IRFs scale linearly with size of shock in single-product model, not in ours

Output Response on Impact



Non-Linear Phillips Curve



Conclusions

- A major shortcomings in standard menu cost models
 - no comovement between inflation and frequency of adjustment
 - See also companion paper with U.S. time series
- Proposed simple extension to remedy these
 - less misallocation from markup dispersion inside the firm
- Model reproduces micro price statistics with much smaller menu costs
- Predicts much more non-linear output responses to monetary shocks